

$$C \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = 2 \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + 3 \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + 5 \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} 2a_1 + 3b_1 + 5c_1 \\ 2a_2 + 3b_2 + 5c_2 \end{bmatrix} \quad (\text{By scalar multiplication and vector addition})$$

$$\therefore CA = \begin{bmatrix} a_1 + b_1 + c_1 & 2a_1 + 3b_1 + 5c_1 \\ a_2 + b_2 + c_2 & 2a_2 + 3b_2 + 5c_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$a_1 = 1, b_1 = 1, c_1 = -1$ will satisfy Row 1
 $a_2 = -1, b_2 = 1, c_2 = 0$ will satisfy Row 2

$$\therefore C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix} \quad AC = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) & 1 \cdot 1 + 2 \cdot 1 & 1 \cdot (-1) + 2 \cdot 0 \\ 1 \cdot 1 + 3 \cdot (-1) & 1 \cdot 1 + 3 \cdot 1 & 1 \cdot (-1) + 3 \cdot 0 \\ 1 \cdot 1 + 5 \cdot (-1) & 1 \cdot 1 + 5 \cdot 1 & 1 \cdot (-1) + 5 \cdot 0 \end{bmatrix}$$

(Row-Column rule of Matrix mult.)

$$= \begin{bmatrix} -1 & 3 & -1 \\ -2 & 4 & -1 \\ -4 & 6 & -1 \end{bmatrix} \neq I_3 \quad \square$$

38) Let $A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$. Construct a 4×2 matrix D using only 1 and 0 as entries, such that $AD = I_2$. Is it possible that $CA = I_4$ for some 4×2 matrix C ? Why or why not?

Solution: Let, $D = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \\ a_4 & b_4 \end{bmatrix}$. $AD = \begin{bmatrix} Ad_1 & Ad_2 \end{bmatrix}$ (By Defn. of matrix multiplication)

$$Ad_1 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_4 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 + a_2 + a_3 \\ a_2 + a_3 + a_4 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} a_1 + a_2 + a_3 & b_1 + b_2 + b_3 \\ a_2 + a_3 + a_4 & b_2 + b_3 + b_4 \end{bmatrix} \quad (\text{By scalar multiplication and vector addition})$$

$$Ad_2 = \begin{bmatrix} b_1 + b_2 + b_3 \\ b_2 + b_3 + b_4 \end{bmatrix} \quad (\text{Similarly}). \quad \therefore AD = \begin{bmatrix} a_1 + a_2 + a_3 & b_1 + b_2 + b_3 \\ a_2 + a_3 + a_4 & b_2 + b_3 + b_4 \end{bmatrix}$$

$$x_2 + x_3 = 1$$